

# T-030 / R-040: the true sphere L-dependence of the few-circles additive

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

**Status.** The bound  $E_+(Q) \leq (2p_{\max} + 1)N^2$  is PROVED for parallel circles with pairwise DISTINCT radii (T7 within that class). For genuine sphere latitudes it applies to subfamilies with  $z_i^2$  pairwise distinct (on a sphere  $z$  and  $-z$  share a radius, so the full latitude union is not automatically distinct-radius). The broader sphere-latitude conjecture  $E_+(Q) \lesssim N^2 \log^{O(1)} N$  is STRONG EVIDENCE (T4), not a theorem; the test radii  $5, 10, \dots, 5L$  are a distinct-radius test family, not an arithmetic sphere family ( $\rho_i^2 + z_i^2 = R$ ). The remaining obstruction is the off-diagonal four-circle in-plane incidence energy, to be attacked in R-033.

R-039 proved  $E_+ \leq 6(L + 1)N^2$  (linear in the circle-cover number  $L$ ) on a sphere. The question (operator direction) is the *true*  $L$ -dependence: is  $E_+ = O(N^2 \text{polylog})$  for circle unions, or is the linear  $L$  real? We answer it: for distinct-radius parallel circles the linear  $L$  is replaced by the height additive multiplicity  $p_{\max}$  (proved), and empirically the energy is  $O(N^2)$  *regardless* of  $p_{\max}$ ; the linear regime occurs only for *equal* radii (a cylinder, off the sphere). Scope: parallel circles (a common axis; “latitudes”); the H-NONLATTICE-REMAINDER frontier; not load-bearing for the  $C_{\text{full}}$  head.

## 2. The refined bound (proved): $p_{\max}$ replaces $L$

**Theorem 1** (height-multiplicity bound, distinct-radius parallel circles). *Let  $Q$  be a union of  $L$  circles in parallel planes (heights  $z_1, \dots, z_L$ ) with distinct radii  $\rho_i$ ,  $N = |Q|$ . Then*

$$E_+(Q) \leq (2p_{\max} + 1)N^2, \quad p_{\max} := \max_H \#\{(i, j) : z_i + z_j = H\}.$$

*Proof.* Write  $m = (M_{xy}, M_z)$ ,  $a = (c_a, z_a)$ ,  $b = (c_b, z_b)$ . Then  $a + b = m$  iff  $z_a + z_b = M_z$  and  $c_a + c_b = M_{xy}$ . Grouping by the planes  $(i, j)$  of  $(a, b)$ ,  $r(m) = \sum_{(i, j): z_i + z_j = M_z} r^{ij}(M_{xy})$ ,  $r^{ij}(M_{xy}) = \#\{(c_a, c_b) \in C_i \times C_j : c_a + c_b = M_{xy}\}$ . For fixed  $M_{xy}$ ,  $c_a \in C_i \cap (M_{xy} - C_j)$ : two circles of radii  $\rho_i, \rho_j$ . For  $i \neq j$  (distinct radii) they are distinct, so  $r^{ij}(M_{xy}) \leq 2$  for *all*  $M_{xy}$ , and  $r^{ij}(0) = 0$  (an antipode of a radius- $\rho_i$  point has radius  $\rho_i \neq \rho_j$ ). For  $i = j$ :  $r^{ii}(M_{xy}) \leq 2$  for  $M_{xy} \neq 0$  and  $r^{ii}(0) = \nu_i$ . Let  $p(M_z) = \#\{(i, j) : z_i + z_j = M_z\}$ . Then for  $M_{xy} \neq 0$ ,  $r(m) \leq 2p(M_z)$ , and

$$\sum_{M_{xy} \neq 0} r(m)^2 \leq 2p(M_z) \sum_{M_{xy} \neq 0} r(m) = 2p(M_z) \sum_{(i, j) \in P(M_z)} \nu_i \nu_j,$$

so summing over  $M_z$  gives  $\leq 2p_{\max} \sum_{i, j} \nu_i \nu_j = 2p_{\max} N^2$ . The  $M_{xy} = 0$  slice receives only the diagonal  $i = j$  (distinct radii kill  $i \neq j$ ):  $r((0, 2z_i)) = \nu_i$ , so  $\sum_{M_z} r((0, M_z))^2 = \sum_i \nu_i^2 \leq N^2$ . Adding,  $E_+ \leq (2p_{\max} + 1)N^2$ .  $\square$

This is the multi-circle refinement of Lemma A (R-025, the single- $T'$  case) and strictly sharper than R-039's  $6(L + 1)N^2$  whenever  $p_{\max} < 3(L + 1)/(2) - \frac{1}{2}$ , in particular for sub-AP height structure (machine: assert A5,  $p_{\max} \leq 2$  for Sidon heights).

### 3. The empirical truth: distinct radii force $O(N^2)$

The bound's  $p_{\max}$  is itself loose. With distinct radii (the sphere case: latitudes have radii  $\sqrt{R - z_i^2}$  – equal for  $z_i$  and  $-z_i$ , hence distinct ONLY when the  $z_i^2$  are pairwise distinct), the  $p(M_z)$  colliding plane-pairs land on *different* in-plane sums  $M_{xy}$  (distinct radii give distinct Minkowski annuli), so the energy never concentrates. Exact integer experiments (distinct rich radii 5, 10,  $\dots$ ,  $5L$ ; assert A2, A3):

| $L$                       | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 10   | 12   |
|---------------------------|------|------|------|------|------|------|------|------|------|
| Sidon heights: $p_{\max}$ | 2    | 2    | 2    | 2    | 2    | 2    | 2    | 2    | 2    |
| $E_+/N^2$                 | 2.60 | 2.45 | 2.38 | 2.43 | 2.38 | 2.33 | 2.29 | 2.26 | 2.22 |
| AP heights: $p_{\max}$    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 10   | 12   |
| $E_+/N^2$                 | 2.60 | 2.48 | 2.52 | 2.70 | 2.78 | 2.72 | 2.74 | 2.77 | 2.83 |

Both stay  $\sim 3$ , *independent of  $L$  and of  $p_{\max}$* : distinct radii suppress to  $O(N^2)$  even at maximal height multiplicity. The contrast is decisive (assert A4): *equal*-radius circles (a cylinder) at AP heights give  $E_+/N^2 = 4.3, 6.0, 7.8, \dots, 22.9$  for  $L = 2, \dots, 12$  with  $E_+/(LN^2) \approx 1.9$  constant – i.e.  $E_+ \sim 2LN^2$ , linear. The product structure  $C \times \text{AP}$  (equal radius makes every in-plane sum coincide) is what creates the linear  $L$ ; the full cylinder mechanism (all  $L$  circles sharing ONE radius) cannot occur for genuine sphere latitudes, where at most two latitudes ( $\pm z$ ) share a radius; the tested linear mechanism is thus absent for sphere latitude subfamilies with  $z_i^2$  distinct.

### 4. Conjecture and the residual (an incidence problem $\rightarrow$ R-033)

**Conjecture (sphere circle-union energy).** For  $Q \subset S^2$  a union of  $L$  latitude circles,  $E_+(Q) \lesssim N^2 \log^{O(1)} N$ ,  $L$ -independent. The evidence is the data of Sec. 3 (STRONG EVIDENCE T4). The residual obstruction to a proof is the *off-diagonal 4-circle in-plane energy*  $\#\{c_a + c_b = c_c + c_d : c_a \in C_i, c_b \in C_j, c_c \in C_k, c_d \in C_l\}$  for distinct-radius circles: the diagonal  $(i, j) = (k, l)$  already gives  $O(N^2)$  (R-021 in-plane), and bounding the genuinely-distinct 4-tuple contribution by  $O(N^2 \text{polylog})$  is a circle-incidence statement – exactly the technology that R-033 (the  $N^{9/4}$  incidence route) addresses. **Falsification gate (pre-registered):** a distinct-radius parallel-circle family with  $E_+/N^2 \rightarrow \infty$  as  $L \rightarrow \infty$  at fixed per-circle richness would refute the conjecture; the scan to  $L = 12$  shows no growth (worst 2.83).

### 5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- ( $\alpha$ ) “The refined bound is just R-039/Lemma A with  $p_{\max}$ .” VALID: it is the height-multiplicity refinement (the  $i \neq j$  in-plane term is  $\leq 2$  by distinct radii). The new content is (i) the clean  $(2p_{\max} + 1)N^2$  form, (ii) the empirical  $O(N^2)$  truth and (iii) the equal-vs-distinct-radius dichotomy that explains where the linear  $L$  comes from.
- ( $\beta$ ) “AP-distinct  $O(N^2)$  is a small- $L$  illusion.” ADDRESSED: tested to  $L = 12$  ( $p_{\max} = 12$ );  $E_+/N^2$  flat at 2.6–2.83. The equal-radius cylinder over the SAME  $L$ -range grows to 22.9, so the flatness is a genuine distinct-radius effect, not a small- $L$  artefact.
- ( $\gamma$ ) “ $O(N^2)$  is asserted, not proved.” UPHELD: the conjecture is STRONG EVIDENCE, not a theorem; the proved statement is only  $(2p_{\max} + 1)N^2$ . The residual is named (off-diagonal 4-circle in-plane energy) and routed to the incidence programme R-033.

( $\delta$ ) *Prediction-correction record.* The first build predicted AP heights  $\Rightarrow$  linear  $E_+$ ; the data refuted it for distinct radii. The note records the correction rather than hiding it; the surviving claim is exactly what the data supports.

**Quantitative sanity check (mandatory).** *Bound.*  $E_+ \leq (2p_{\max} + 1)N^2$  on all distinct-radius configs (assert A1). *Dichotomy.* distinct-radius AP  $E_+/N^2 \leq 2.83$  vs equal-radius AP  $E_+/N^2 = 22.9$  at  $L = 12$  – a  $8\times$  separation isolating the equal-radius mechanism. *Sharper.*  $(2p_{\max} + 1)N^2 < 6(L + 1)N^2$  for Sidon,  $L \geq 3$  (assert A5). *Estimator.* vectorised  $E_+ =$  brute  $O(N^4)$  (396, assert A6). *Reproducibility.* deterministic exact-integer; 6/6 asserts, exit 0.

## 6. Result footer

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Result ID:              | R-040 (true sphere L-dependence; height-multiplicity refinement) / T-030 frontier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Precise statement:      | (proved) L parallel circles, DISTINCT radii, heights $z_i \Rightarrow E_+(Q) \leq (2 p_{\max} + 1) N^2$ , $p_{\max} = \max_H \#\{(i,j): z_i+z_j=H\}$ . Replaces R-039's linear L by the height additive multiplicity. (empirical) distinct radii (distinct-radius parallel circles; sphere latitudes with $z_i^2$ pairwise distinct) $\Rightarrow E_+/N^2$ BOUNDED ( $\sim 3$ ) even at AP heights ( $p_{\max} \sim L$ ): $E_+ = O(N^2)$ , L-independent. The linear-L regime is an EQUAL-radius (cylinder, off-sphere) artifact ( $E_+ \sim 2 L N^2$ , product $C \times$ AP). |
| Scope:                  | parallel circles (latitudes); T-030 ( $Q$ in $S^2$ ; H-NONLATTICE-REMAINDER). Not load-bearing for C_full head.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Evidence grade:         | Refined bound $(2 p_{\max}+1)N^2$ : PROVED (T7-within-class, elementary; distinct radii essential). Conjecture $E_+=O(N^2 \text{ polylog})$ for sphere latitude unions: STRONG EVIDENCE (T4), scan to $L=12$ no growth. Does NOT close T-030: arbitrary- $Q N^{\{2+\epsilon\}}$ STILL OPEN (record $N^{\{9/4\}}$ , R-033).                                                                                                                                                                                                                                                      |
| Reproduction command:   | python3 codes/vacuum/dr2_t030_height_multiplicity.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Expected output:        | 6/6 asserts PASS, exit 0; JSON at claims/<br>B5-BEYOND-LAYER-BOUND/runs/260614-dr2-t030-height-multiplicity/result.json                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Falsifier:              | a distinct-radius parallel-circle family with $E_+/N^2 \rightarrow$ infinity as L grows at fixed per-circle richness (refutes the $O(N^2)$ conjecture); a distinct-radius config with $E_+ > (2 p_{\max}+1) N^2$ (refutes the theorem). Both machine-checked false to $L=12$ .                                                                                                                                                                                                                                                                                                  |
| Tier before / after:    | NONE -- operator decides. No gate/tier/hypothesis flip. B5 stays T7-SCOPE_{admissibility-bounded} given A1; T-030 OPEN.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| No-overclaim statement: | The $O(N^2)$ sphere behaviour is STRONG EVIDENCE, not a theorem; only $(2 p_{\max}+1)N^2$ is proved. The residual (off-diagonal 4-circle in-plane energy) is an incidence problem routed to R-033. Distinct radii are essential; the cylinder (equal radii) genuinely is linear.                                                                                                                                                                                                                                                                                                |
| Next required action:   | Proceed to R-033: bound the off-diagonal 4-circle in-plane energy / improve circle incidences below $N^{\{9/4\}}$ , which would both prove the $O(N^2)$ sphere conjecture and push the arbitrary- $Q$ T-030 record.                                                                                                                                                                                                                                                                                                                                                             |